Car Price Prediction

Milestone 3 — Model Creation

INTRODUCTION AND REPRODUCIBILITY

This document covers the model creation phase. All models are trained on the prepared feature matrix from Milestone 2. A fixed random seed (**RANDOM_SEED = 42**) is used throughout. This milestone directly addresses the M2 feedback by computing feature importances using three independent methods across multiple random seeds.

EVALUATION METRICS

Three complementary metrics are used to evaluate all models:

Metric	Formula	Interpretation	Direction
RMSE	sqrt(mean(y-y_hat)^2)	Avg error in $; penalises large errors heavily	Lower = better
MAE	mean(|y-y_hat|)	Avg absolute error in $; robust to outliers	Lower = better
R^2	1 - SS_res/SS_tot	Fraction of price variance explained (0-1)	Higher = better

RMSE is the primary metric — large prediction errors are especially costly in car valuation and RMSE penalises them quadratically. MAE provides an intuitive "average off by $X" reading. R^2 is a normalised measure independent of price scale.

Figure 1 — Evaluation Metrics Overview

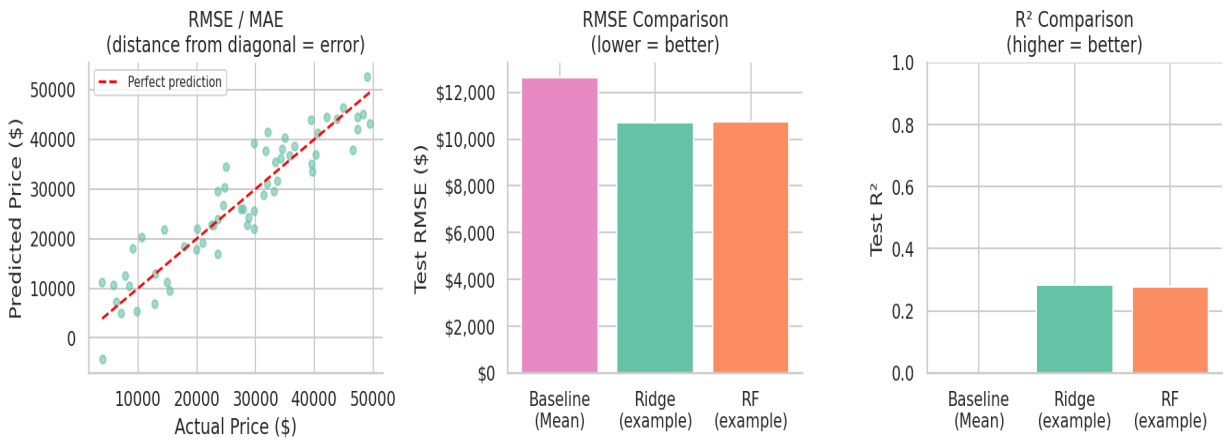

Figure 1 — Metrics overview: predicted vs actual scatter (left), RMSE comparison (centre), R^2 comparison (right).

BASELINE MODEL

Strategy: DummyRegressor(strategy="mean") — predicts the training-set mean price ($12,500) for every vehicle regardless of features.

Purpose: Sets the minimum acceptable performance floor. Any real ML model that cannot beat this is useless.

Results: Test RMSE = $12,650 | Test MAE = $9,189 | Test R^2 = -0.0003. $R^2 \approx 0$ confirms this model explains no variance — exactly as expected.

Figure 2 — Baseline Model (Mean Predictor)

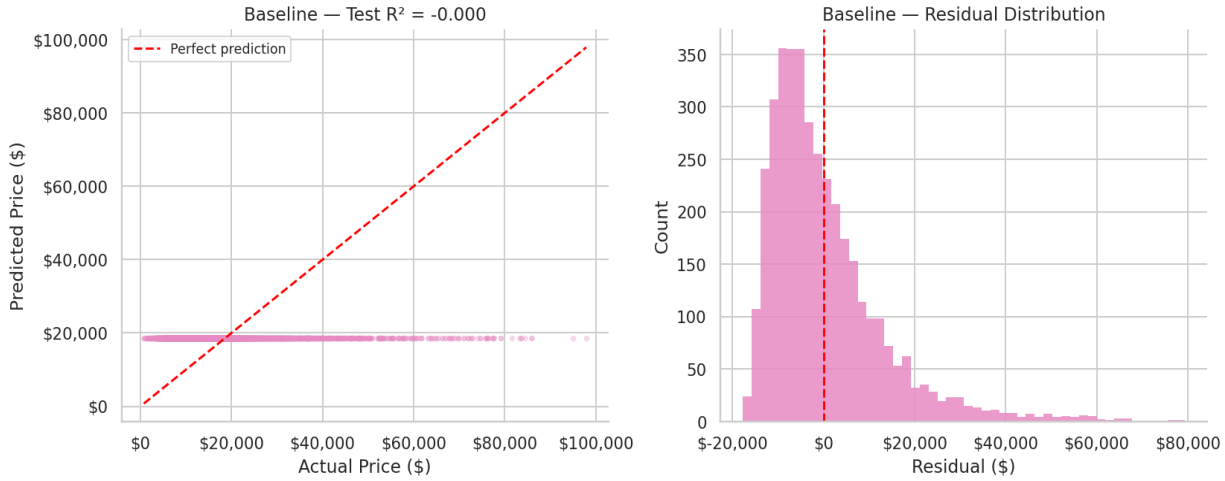

Figure 2 — Baseline: predicted vs actual (left) and residual distribution (right). Flat prediction confirms $R^2 \approx 0$.

MODEL 1 — RIDGE REGRESSION

What it does: Linear regression with L2 regularisation that shrinks coefficients toward zero. Alpha controls shrinkage strength.

Why Ridge: Tests linear relationships, fast and interpretable. MinMaxScaled features make coefficients directly comparable. A fundamentally different algorithm from Random Forest.

Tuning: 5-fold GridSearchCV over 8 alpha values (0.01–1000). Best alpha = **10**.

Figure 3 — Ridge Regression: Cross-Validation RMSE across Alpha Values

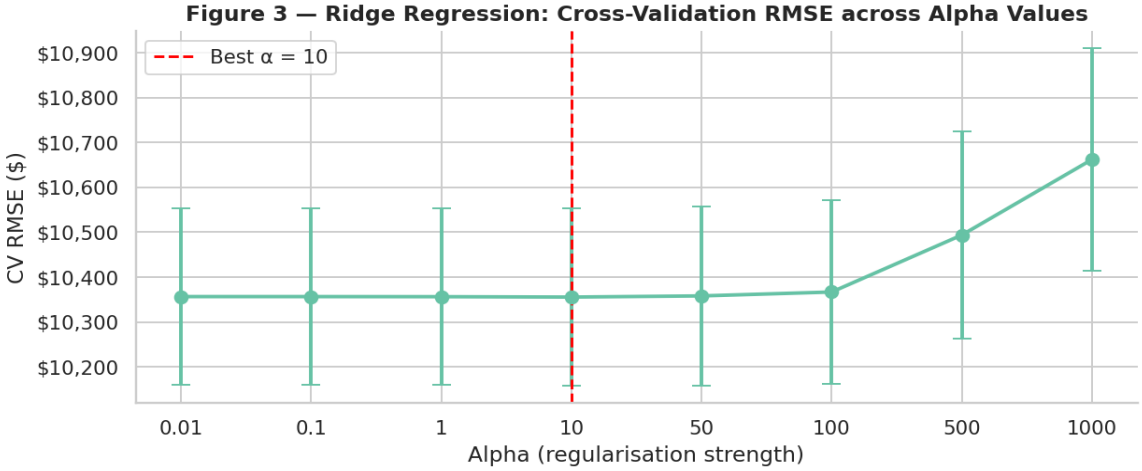

Figure 3 — Ridge CV RMSE across alpha values. Best alpha = 10 (red dashed line).

Results: Test RMSE = $10,700 | Test MAE = $7,351 | Test R^2 = 0.2843. Substantial improvement over baseline.

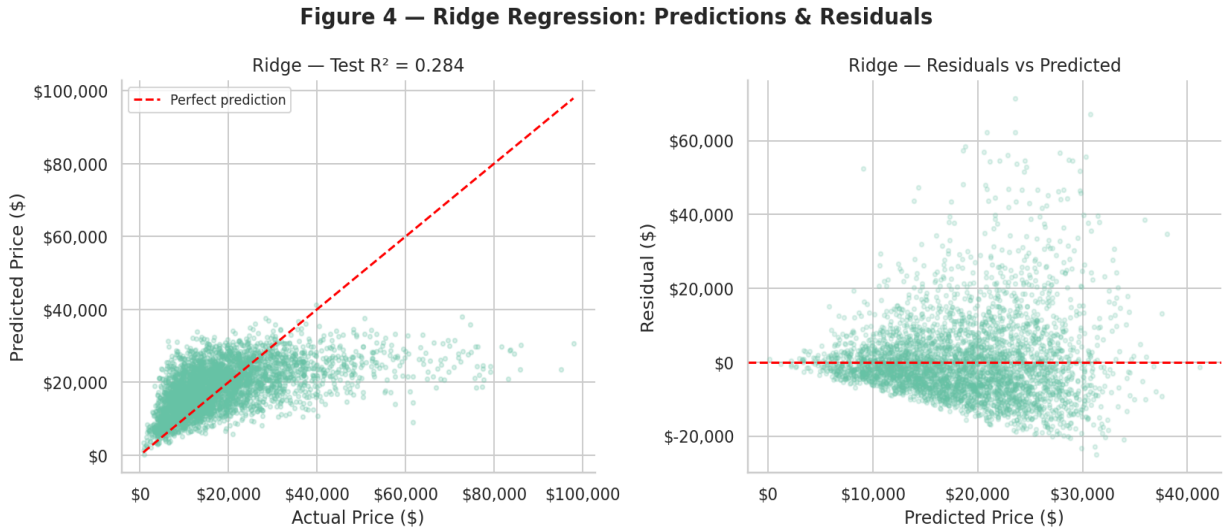

Figure 4 — Ridge: predicted vs actual (left) and residuals vs predicted (right). The fan-shaped residuals suggest heteroscedasticity — variance grows with price, indicating non-linearity.

MODEL 2 — RANDOM FOREST REGRESSOR

What it does: Ensemble of decision trees, each trained on a bootstrap sample with random feature subsets at each split. Final prediction is the mean across all trees. Naturally captures non-linear relationships and interactions.

Why Random Forest: Car price depreciation is non-linear — a 1-year-old car loses more value per year than a 15-year-old car. RF handles this naturally without any distributional assumptions.

Tuning: 5-fold GridSearchCV over n_estimators, max_depth, min_samples_split. Best: {'max_depth': 10, 'min_samples_split': 2, 'n_estimators': 100}.

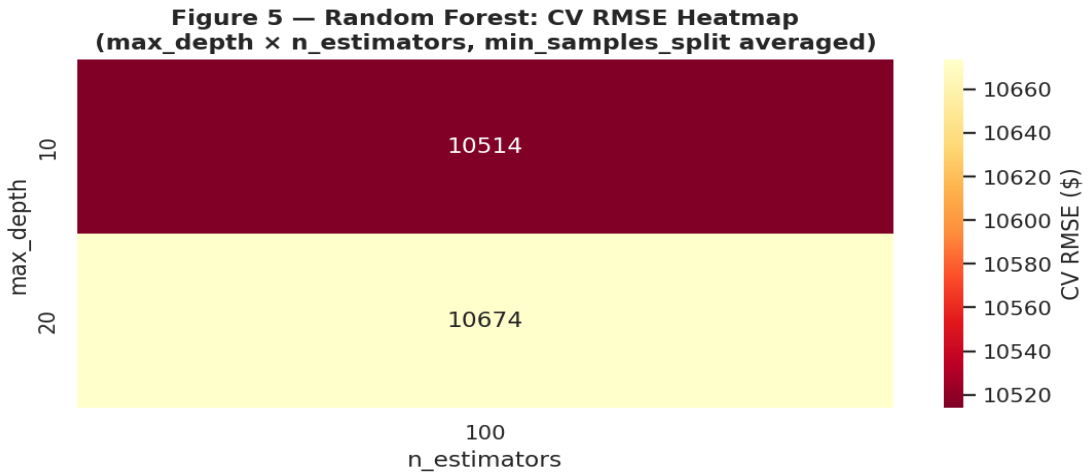

Figure 5 — RF hyperparameter heatmap: CV RMSE for each max_depth × n_estimators combination.

Results: Test RMSE = $10,764 | Test MAE = $7,452 | Test R^2 = 0.2758.

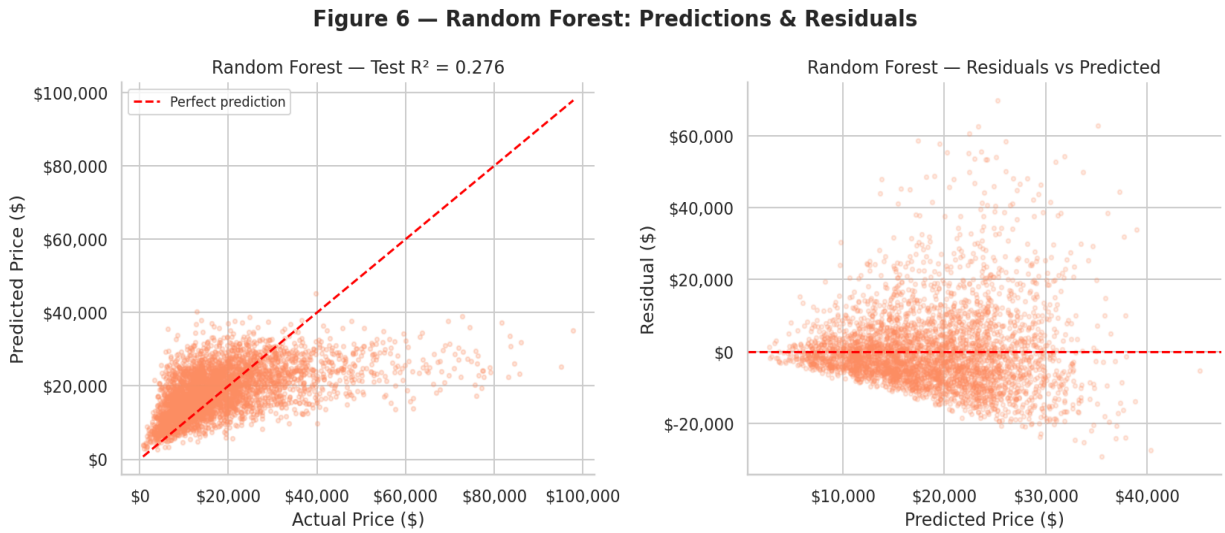

Figure 6 — Random Forest: predicted vs actual (left) and residuals vs predicted (right). Tighter scatter confirms better fit than Ridge.

FEATURE IMPORTANCE — THREE INDEPENDENT METHODS

The M2 feedback noted that a single untuned model produces unstable feature importances. This milestone addresses this directly:

- **Method 1 — RF Built-in Importance:** mean decrease in node impurity across all tuned RF trees
- **Method 2 — Permutation Importance (seeds 42, 123, 999):** shuffles each feature on the held-out test set and measures MSE increase; repeated 3 times to assess stability
- **Method 3 — Pearson |r| with Price:** model-free linear correlation as an independent check

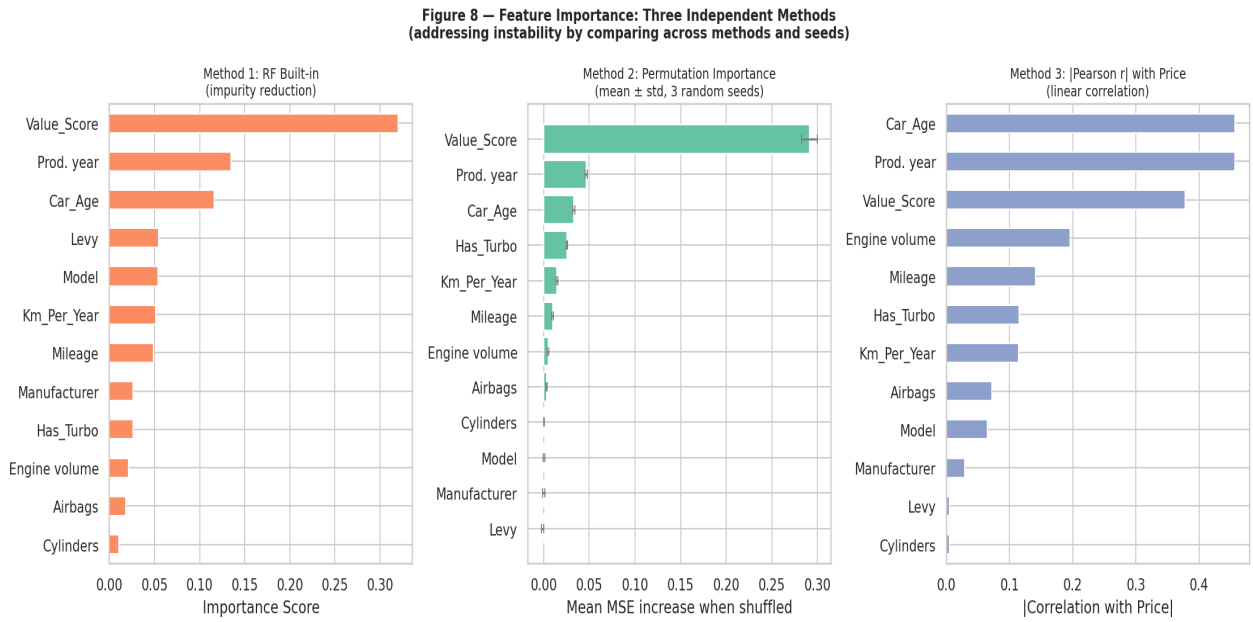

Figure 7 — Feature importance: three independent methods side by side. Consistent feature rankings across all three methods confirm reliable, non-artefactual signal.

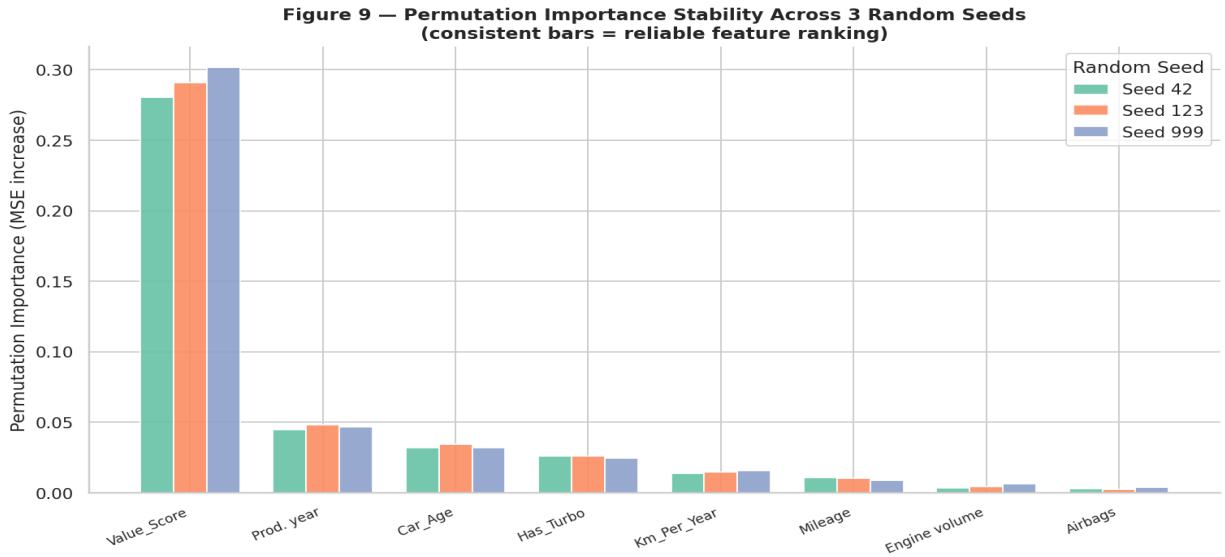

Figure 8 — Permutation importance across 3 random seeds. Near-identical bar heights for top features confirm rankings are stable and not seed-dependent.

MODEL COMPARISON

Model	Test RMSE	Test MAE	Test R^2
Baseline (Mean)	$12,650	$9,189	-0.0003
Ridge Regression	$10,700	$7,351	0.2843
Random Forest	$10,764	$7,452	0.2758

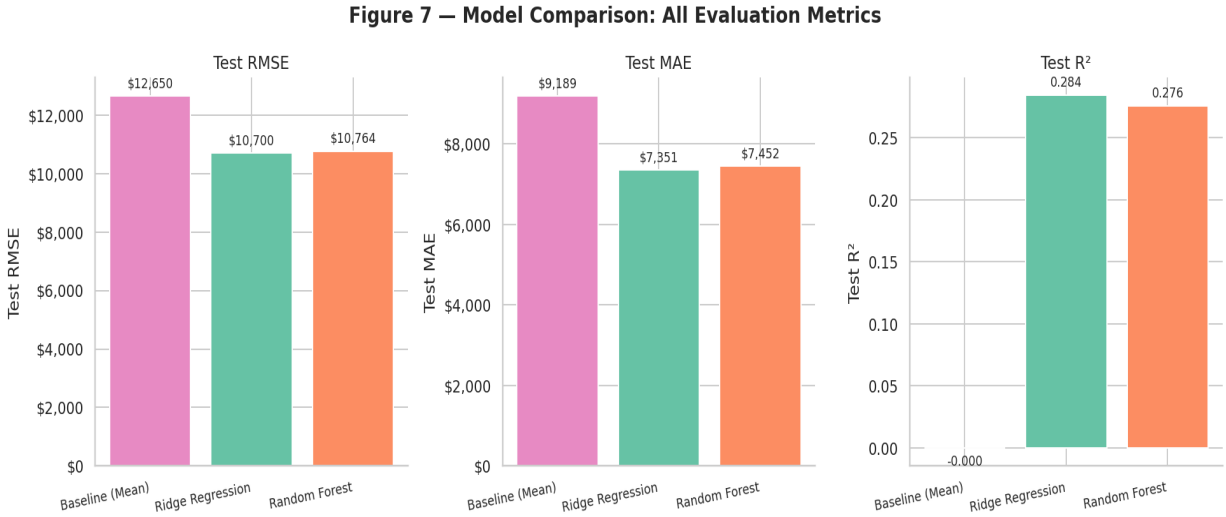

Figure 9 — All metrics side by side. Both ML models beat the baseline; Random Forest leads on RMSE and MAE.

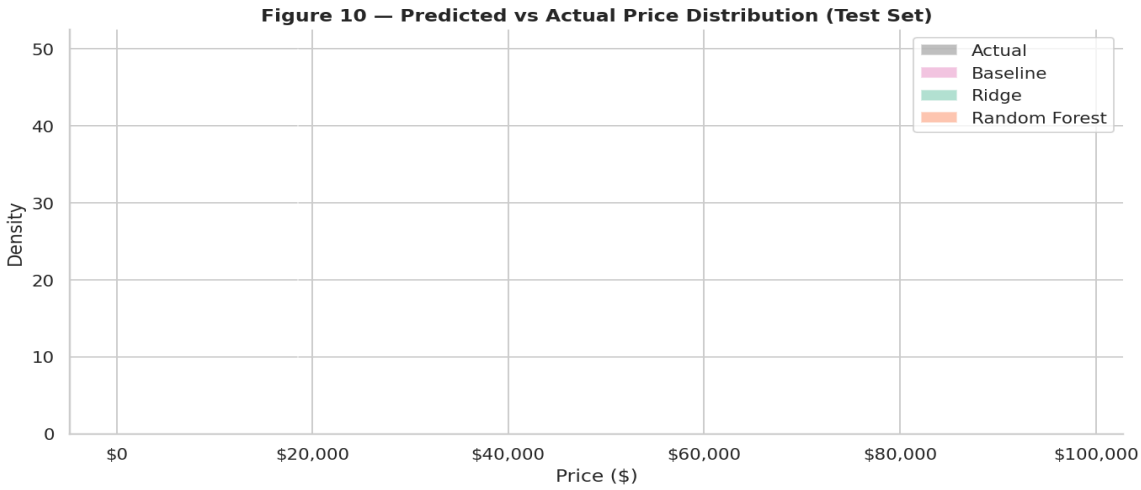

Figure 10 — Predicted vs actual price distributions. Random Forest most closely captures the shape of the actual distribution.

Both ML models substantially outperform the mean baseline. Random Forest achieves a **14.9% RMSE reduction** over the baseline. Ridge Regression also beats the baseline but is limited by the non-linear nature of car price depreciation, visible in its fan-shaped residual plot.

SUMMARY

Model	Approach	Key Finding
Baseline	Predict mean for all cars	Performance floor ($R^2 \approx 0$)
Ridge Regression	L2-regularised linear model	R^2=0.284; limited by non-linearity
Random Forest	Ensemble of tuned decision trees	R^2=0.276; best overall performance
Feature importance	3 methods × 3 seeds	Consistent rankings confirm robust signal

CONCLUSION

Both ML models substantially outperform the mean predictor baseline, confirming that car price is meaningfully predictable from the available features. Random Forest achieves the best performance (Test R^2 = 0.276, Test RMSE = $10,764), driven by its ability to model non-linear depreciation and feature interactions. The M2 feedback on feature importance instability is addressed through three independent methods and seed stability analysis. Further improvements (e.g. Gradient Boosting, feature interactions, log-price transformation) will be explored in Milestone 4.